

# Variance swap. Trading commodity volatility directly, no option in between.

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LinkedIn post snapshot

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Most vol exposure in commodity gets expressed through options. Buy a straddle, sit short, delta-hedge, collect the difference between implied and realized. It works but it's noisy. Strike selection, time decay, delta hedging cost all leak P&L away from the pure vol view. The variance swap solves that. It's a direct trade on realized variance.

The 5 things to understand about variance swaps in commodity.

## 1. What it is

A swap where one leg pays the realized variance of the underlying over a period, the other pays a fixed strike. If realized variance exceeds the strike, the long-variance side wins. If realized variance falls below, the short side wins. No options involved. No delta hedging required.

## 2. Why it's cleaner than a straddle

A straddle is a vega bet contaminated by gamma, theta, skew dynamics, and the cost of delta hedging. Variance swap isolates the variance exposure exactly. No strike to choose. No delta to hedge. The P&L is a direct function of how much the underlying moved, period.

## 3. Where it has liquidity in commodity

Crude oil (Brent and WTI), natural gas (Henry Hub and TTF). Limited but real liquidity in gold and copper. Almost nothing in agricultural commodities, where vol exposure stays in straddle form. Growth has concentrated in deep, liquid energy markets where vol harvest strategies have institutional appetite.

## 4. The tail risk

Variance is calculated as the sum of squared returns. A single large move contributes disproportionately. When a tail event hits, realized variance can blow far past the strike. Short variance positions have unbounded P&L exposure to a single large move. 2020 oil and the 2022 Russia gas dislocation both produced variance swap losses that exceeded any reasonable expectation.

## 5. Who uses them

Institutional vol sellers run them as alternative to listed straddle strategies. Some hedge funds use them as macro vol expression on energy. Corporates rarely touch them because the operational complexity exceeds the benefit. The structure exists for the trading floor, not the treasury floor.

Variance swap is the cleanest instrument to express a pure vol view. It also has the most asymmetric tail. The clean vega exposure cuts both ways: there's nothing else in the P&L to dilute a bad outcome. If you trade vol systematically, the structure is worth knowing. If you trade vol opportunistically, options usually serve you better.

For anyone using variance swaps in commodity: where do you set the tail cap, and how do you size around it?

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### **Selected comments**

**Eric Barthe** (Founder, The Derivatives Bootcamp; Ex-Goldman Sachs Exotic Trader): "As the name says, the variance swap is the cleanest instrument to express a pure variance view, but not a pure vol view."

**Taylor Lukof** (Founder & CEO, ABR Dynamic Funds): "Be careful introducing counterparty risk in a tail-event product... worst thing would be to not get paid your insurance when you need it most!"

**Xiaoou John Cao** (Market Risk): "I like this visual, but perhaps could add more details on who are the sellers and buyers and which tenor or underlying is the most popular/liquid ones...?"

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